nature communications

Article

https://doi.org/10.1038/s41467-026-69142-4

Experimentally separating vacuum fluctuations from source radiation

Received: 26 March 2025

Accepted: 27 January 2026

Published online: 17 February 2026

Check for updates

Alexa Herter[1] ✉, Frieder Lindel[2,3,4], Laura Gabriel[1], Stefan Yoshi Buhmann[5] & Jérôme Faist[1] ✉

The distinction between vacuum-field and source-radiation effects in phenomena such as the Lamb shift, Casimir forces, and spontaneous emission relies on additional assumptions at the theoretical level, and an experimental approach was never considered feasible. Fermi's two-atom problem, a Gedankenexperiment on how atoms interact with the electromagnetic field via vacuum and source radiation, provides key theoretical insight. Advances in ultrafast optics now enable experimental analogues using two laser pulses in a nonlinear crystal. Here we demonstrate the detection of vacuum- and source-radiation-induced correlations between two pulses, separated by their causal properties. Specifically, vacuum fluctuations and source radiation are shown to correlate distinct quadratures of near-infrared pulses, enabling individual probing via phase-sensitive detection. Our results experimentally verify the time-domain fluctuation-dissipation theorem at the quantum level and open avenues for studying quantum radiation effects in time-dependent media, including entanglement harvesting from the vacuum and quantum field detection in curved-space analogues.

Fermi's two-atom setup consists of two atoms in empty space whose interaction with the vacuum radiation field is abruptly switched on. Studying the subsequent build-up of correlations between the atoms reveals fundamental properties of quantum fields in space and time[1–4]. In the Heisenberg picture, independent of the initial state of the two atoms, two different terms contribute to these correlations[5–8], as illustrated in Fig. 1a. On the one hand, correlations between the two atoms arise from their individual interaction with the ground-state electromagnetic field—the *vacuum fluctuations*—whereby correlations intrinsically existing in the vacuum field are swapped to the atoms[1–4]. On the other hand, the two atoms can exchange a photon. The emission of this photon is triggered by the interaction of one atom with its own radiative field (radiation reaction). This radiation is known as *source radiation*[1,6] and interacts with the other atom, which leads to correlations between the atoms. Mathematically, individual contributions of vacuum fluctuations and source radiation to fundamental processes in quantum electrodynamics, such as spontaneous emission or the Lamb shift, are in general not uniquely determined, since they depend on the initial ordering of the atomic and field operators in the light-matter interaction Hamiltonian[6,9,10].

While for most of the phenomena associated with interactions with the ground state of light, vacuum fluctuations and source radiation can therefore be seen as inseparable "two sides of the same quantum mechanical coin"[10], they have a different character regarding causality in Fermi's two-atom problem: The vacuum-induced correlations arise instantaneously upon turning on the interaction, while the exchange of a photon leads to correlations only after the propagation time $\delta t = \frac{r}{c}$[7,8]. However, different choices of the aforementioned operator ordering can lead to a non-causal description of source radiation, making its interpretation ambiguous[7]. This issue remains purely theoretical unless the two contributions can be observed independently. Yet, an experimental investigation has long been considered unfeasible due to the challenge of initiating the two atoms, rapidly switching their interaction with the field on ultra-short

[1]Institute of Quantum Electronics, ETH Zürich, Zürich, Switzerland. [2]Institute for Theoretical Physics, ETH Zürich, Zürich, Switzerland. [3]Quantum Center, ETH Zürich, Zürich, Switzerland. [4]Departamento de Física Teórica de la Materia Condensada and Condensed Matter Physics Center (IFIMAC), Universidad Autónoma de Madrid, Madrid, Spain. [5]Institut für Physik, Universität Kassel, Kassel, Germany. ✉e-mail: alexa.herter@web.de; jfaist@ethz.ch

https://doi.org/10.1038/s41467-026-69142-4

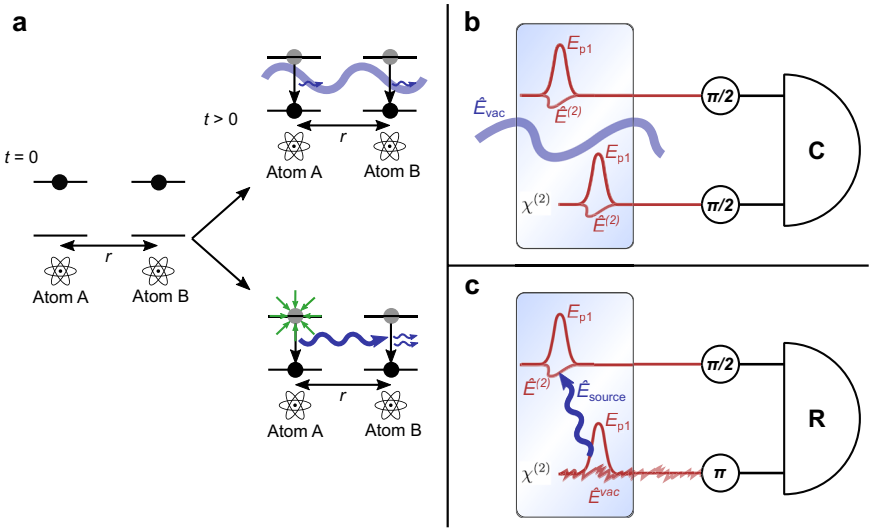


Fig. 1 | Analogy between Fermi's two-atom problem and electro-optic correlations. a In Fermi's two-atom problem, two initially uncorrelated excited atoms A and B are separated by a distance r and their interaction with the electro-magnetic field is turned on at time $t = 0$. The atoms can become correlated either by interacting individually with the surrounding vacuum field fluctuations (upper sketch), or by exchanging source radiation (one atom emitting a photon by interacting with its own radiation field, which then interacts with the other atom) (lower sketch). **b** Similar to the vacuum-induced correlations in Fermi's two-atom setup, the polarization state of two laser pulses can become correlated by interacting with the surrounding vacuum field inside an electro-optic crystal. The generated nonlinear field is detected in a homodyne detection scheme[13,15]. **c** Additionally, one laser pulse can emit a photon due to the interaction with the ground-state fluctuations in its own frequency range and the generated photon can then interact with the second pulse. To probe this process, the phase relation in one of the homodyne detection schemes has to be changed from $\frac{\pi}{2}$ to π[7] (see main text).

timescales, and detecting the resulting correlations[11]. Here, we overcome these limitations using an all-optical version of Fermi's two-atom setup and probe the contributions from source radiation and vacuum field fluctuations, separated by causality, individually for the first time.

In contrast to atoms, the interaction of laser pulses with a surrounding electromagnetic field can be initiated and terminated on ultrashort time scales as the pulses enter and leave a nonlinear material. This unique property has enabled an all-optical analog of Fermi's two-atom setup, where vacuum-induced correlations between atoms are mapped onto two spatially separated laser pulses interacting with the vacuum field inside a nonlinear crystal[12] (see Fig. 1b). The implementation relies on electro-optic sampling, in which near-infrared probe pulses couple to a terahertz or mid-infrared field (possibly in the vacuum state and restricted to the spectral width of the probe), generating a second-order nonlinear field polarized perpendicular to the probe[13–15]. This nonlinear interaction modifies the polarization state of the two probe pulses, directly paralleling the transition of the atoms in the Fermi setup. Here, however, the resonant single-mode transition of two-level atoms is replaced by an off-resonant, broadband coupling of multimode near-infrared pulses to terahertz field modes within the spectral width of the probes. The use of femtosecond pulses thus not only allows the interaction to be switched on and off with high temporal precision, but also enables the application of advanced quantum optics detection techniques to analyze the polarization state of the probes (see refs. 7,12,16 and Supplementary Information Note 1.A for further comparison between Fermi's two-atom setup and two-beam electro-optic sampling).

Recently, an experiment similar to the one in ref. 12 has been theoretically suggested[7], which would be capable of individually detecting correlations caused by source radiation as well (see Fig. 1c), and in the following, we demonstrate its experimental implementation.

Results

The idea relies on the fact that in the electro-optic sampling analog of Fermi's two-atom setup, vacuum-field fluctuations (given by the correlation function of the THz vacuum field $C(\mathbf{r},\mathbf{r}',t,t') = \frac{1}{2}\langle\{\hat{\mathbf{E}}_{vac}(\mathbf{r},t),\hat{\mathbf{E}}_{vac}(\mathbf{r}',t')\}\rangle$) and source radiation (given by the response function $R(\mathbf{r},\mathbf{r}',t,t') = (i/\hbar)\theta(t-t')\langle[\hat{\mathbf{E}}_{vac}(\mathbf{r},t),\hat{\mathbf{E}}_{vac}(\mathbf{r}',t')]\rangle$) correlate different quadratures of the two near-infrared pulses, which allows one to individually probe them by changing the phase plate in the employed homodyne detection scheme (see Fig. 1b, c). This distinction mirrors the situation in Fermi's two-atom problem, where different Pauli matrices are correlated depending on whether the correlation originates from vacuum fluctuations or from source radiation[7].

In the vacuum-field contribution, the near-infrared field of the two pulses individually interacts with the vacuum field at terahertz frequencies. Due to the retardance of $\frac{\pi}{2}$ occurring in the non-linear process, the generated field that contains the information about the THz field is out-of-phase with the probe field. As a consequence, correlations induced by terahertz vacuum fluctuations can be detected through quarter-wave plates in the homodyne detection schemes of both probe beams, as schematically illustrated in Fig. 1b. The instantaneous correlation of the two pulses induced by the terahertz vacuum fluctuations is obtained from the individually detected signals as a function of different time delays between the two probe pulses[12]. In the source radiation contribution schematically depicted in Fig. 1c, the electric field of one probe pulse interacts with the copropagating ground-state fluctuations in the near-infrared, resulting in the emission of source radiation. The second pulse then interacts with that terahertz source radiation in the same way as with the terahertz vacuum field again leading to a nonlinear field contribution that has a quadrature out-of-phase with the probe field. In contrast, to trigger the source radiation, the probe field of the first pulse interacted with the in-phase quadrature of the near-infrared vacuum fluctuations. To observe the correlations resulting from source radiation, one thus has to probe the first pulse with a half-wave plate—thereby accessing the quadrature that is in-phase with the pulse—and the second with a quarter-wave plate as in the vacuum field contribution (see ref. 7 for details and Supplementary Information Note 1.B).

https://doi.org/10.1038/s41467-026-69142-4

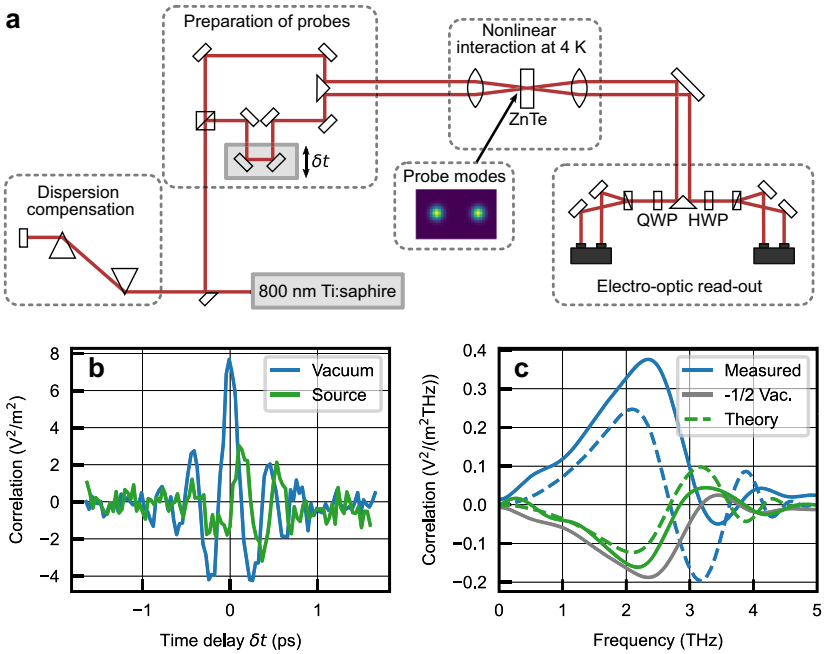


Fig. 2 | Experimentally observed correlations. a Within the optical setup to measure electro-optic correlations, a prism compressor is used to compensate for the dispersion of the transmission optics in the setup to achieve 110 fs long pulses inside the nonlinear crystal. Two individual probe signals are prepared whose time-delay δt is controlled using a delay stage. Inside a cryostat, the probe signals are focused tightly into the zinc-telluride crystal (ZnTe) at 4K. Afterwards the polarization change on both probe beams is individually evaluated on a balanced detector. Depending on the experiment, a quarter-wave plate (QWP) is used to access the change in ellipticity or a half-wave plate (HWP) to observe a rotation in the polarization. **b** Correlations between the signals from the two balanced detectors, which is sensitive to the correlations caused by terahertz vacuum fluctuations (blue, two quarter wave plates in the detection setup) and source radiation (green, one quarter and one half-wave plate in the detection setup), is plotted depending on the time-delay between the two pulses δt. **c** Experimentally observed correlation are shown obtained from the Fourier-transformation of the time-traces shown in (**b**). The vacuum-induced correlation is plotted in blue and green is the imaginary part of the correlation caused by source radiation. The simulated spectral correlations (see "Methods" and ref. 7) are plotted with dashed lines and the vacuum-induced correlation multiplied by a factor of $-\frac{1}{2}$ is plotted in gray. Details about the acquisition and analysis of the experimental data is reported in the Supplementary Note 2.A–C.

For the experimental realization, a similar optical setup as in ref. 12 is utilized and schematically shown in Fig. 2a. To compensate for dispersion and obtain 110 fs-short pulses inside the detection crystal, the probe signal is pre-stretched in a prism compressor before being separated into the two individual probe signals (see the "Methods" section about experimental setup and Supplementary Information Note 2.D). The correlations between different quadratures of the field amplitudes of the two laser pulses emerging from the crystal are detected by probing the correlations between two sets of balanced detectors, each of which has a tunable waveplate determining the probed quadrature (see Fig. 1b, c and the "Methods" section). In agreement with ref. 12, the vacuum-induced correlation measured with two beams at a target distance of 50 µm and with a quarter-wave plate on both detection lines oscillates symmetrically around zero time delay δt with the maximum correlation for $\delta t = 0$ (blue line in Fig. 2b). Compared to previous results, the reduced pulse duration increases the observed peak correlation to more than 5 V^2/m^2[12,15].

For the setup sensitive to correlations stemming from source radiation (combination of a half-wave plate on one and a quarter-wave plate on the other detection line), the temporal correlation (green line in Fig. 1b) is asymmetric. Confirming the causal nature of the correlations induced by source radiation, the signal occurs mainly for positive time delays $\delta t > 0$, while the residual correlations $\delta t \lesssim 0$ arise from the finite spatio-temporal Gaussian extension of the laser pulses[12].

The Fourier transform of the correlation traces for vacuum fluctuations and source radiation are plotted in Fig. 2c. Compared to our previous results, the signal bandwidth now extends to 4.5 THz due to the shorter pulse length[12]. The initial increase in the signal with frequency up to 2 THz is due to the increasing strength of the vacuum fluctuations with frequency. Above 2.5 THz, the two signals change sign as the distance between the two beams becomes comparable to the wavelength inside the crystal. The slight shift of the zero crossing of the experimentally observed vacuum-induced correlation compared to the theoretical model might be caused by a slight shift of the beam distance towards 30 µm for the measurement in a configuration sensitive to vacuum fluctuations (details in the Supplementary Information Note 2.C).

The vacuum field and source radiation signals are further not independent from each other but connected via the fluctuation-dissipation theorem in the quantum limit (zero temperature). It predicts a connection between the correlation that stems from vacuum fluctuations and source radiation in the time domain through the Hilbert transform, which is validated in the experimental data in Fig. 2b through the phase shift of $\frac{\pi}{2}$ between the two correlations[7,17]. In the frequency domain, this leads to a direct relation between the real part of the vacuum-induced correlation (blue solid line) and the imaginary part of the correlation caused by source radiation (green solid line), which are plotted in Fig. 2c. Theoretic calculations (plotted in dashed lines) predict that the imaginary part of the source radiation signal is equivalent to the vacuum-induced correlations multiplied by a factor $-\frac{1}{2}$[7]. When comparing the experimentally observed correlations, the imaginary part of the correlation caused by source radiation (green line) matches well the expectation based on the vacuum-induced correlation (gray line). The slight deviation between those two curves and between the experimental and theoretical data is mainly caused by the distance between the two beams inside the crystal, which deviated from the target value of 50 µm and between the two measurements (details in the Supplementary Information Note II.C).

Discussion

We have presented an individual experimental detection of correlations arising between two laser pulses due to vacuum fluctuations and source radiation naturally separated by their causal properties. Dalibard et al.[6] proposed already in 1982 a way to lift the ambiguity between source radiation and vacuum fluctuations that is consistent with ours by demanding the operators describing the two contributions to be each Hermitian and therefore physical observables. Our results hence confirm the physical meaning of such a separation and support Dalibard's argument. While causality enables the separation of vacuum fluctuations and source radiation in Fermi's two-atom problem, Dalibard's method, now validated by our experiment, provides a general approach applicable to any effect involving these fundamental quantum contributions.

Furthermore, the measurements agree with the predictions of the time-domain fluctuation-dissipation theorem as expected from the discussion of the two-atom problem[7]. The direct connection of the spectral correlations induced by source radiation and vacuum fluctuations offers a new access to vacuum fluctuations through the measurement of source radiation and vice versa.

So far the experiment has been performed at 4K to suppress any thermal radiation in the terahertz regime. While prior studies have demonstrated the electro-optic correlation induced by thermal photons at room temperature[15,18], the correlation caused by source radiation is anticipated to be independent of the optical state in the THz regime[7,8]. Consequently, the configuration that has been demonstrated, which utilizes a half-wave plate and a quarter-wave plate, possesses the capability to detect the correlations caused by a single THz photon within a thermal photon pool at room temperature.

Furthermore, the experiment demonstrates time-domain access to quantum fields at the single photon level. This offers new experimental possibilities for the study of quantum radiation phenomena in time-varying media[19], such as the dynamical Casimir effect[20], or in relativistic quantum information theory[21,22], e.g., for the experimental demonstration of entanglement harvesting from the vacuum state[23] or for probing quantum correlations of relativistic fields in analog curved space-times[24], including event horizons[25].

Methods

Experimental setup

The optical setup to detect electro-optic correlations is shown schematically in Fig. 2a of the main text. We are using the pulsed signal of a titanium-sapphire laser centered at 800 nm to generate the probe signal. To compensate for the dispersion caused by several transmission optics, we implemented a prism compressor. The beam position is actively stabilized to reduce long-term drifts. A beam splitter divides the intensity of the pulsed signal equally into two beam paths, and a delay stage controls the time delay δt between the two beams. The two beams are placed in close proximity to each other, and two piezo-controlled mirrors set the angle between the two beams. Afterwards, the two beams are directed inside a dilution fridge.

Inside the cryostat, a 50 mm lens focuses the beams into a ⟨110⟩-cut zinc-telluride crystal. The angle of the two beams determines the distance of the two beams in the focal plane and is adjusted to a distance of 50 µm in the current experiment. Since the beam waist measures 10 µm, the two beams are well-separated inside the nonlinear crystal. A second lens re-collimates the beams afterwards. Both lenses and the nonlinear crystal are thermally connected to the 4K-plate of the cryostat to suppress thermal radiation in the terahertz regime. The two beams are propagating along the $\langle\bar{1}\bar{1}0\rangle$-axis of the zinc telluride and are polarized along the ⟨001⟩-direction.

After leaving the cryostat, the two probe beams are separated, and the change in polarization is analyzed individually using the combination of a wave plate, a polarizing beam splitter, and a balanced detector. The correlation of the two signals is calculated for each individual pulse pair using a fast acquisition card. Further details about the setup and data acquisition, which is suppressing slowly varying correlations, are reported in ref. 15.

Experimentally observed correlation

In general, the signal observed in electro-optic sampling can be expressed in terms of a difference in photon-number $\hat{S} = \hat{N}_s - \hat{N}_p$ of the two outputs of the polarizing beam splitter (perpendicular $\hat{N}_s$ and parallel $\hat{N}_p$ polarized) measured by the balanced detector[13]. For a wave plate with arbitrary retardance γ, the signal is given by:

$$\hat{S}_\gamma = 4\pi\epsilon_0 c \int \mathrm{d}^2 r_\perp \int_0^\infty \mathrm{d}\omega \, \frac{n(\omega)}{\hbar\omega} \left(P(\gamma)\hat{E}_x^\dagger(\vec{r}_\perp,\omega)\hat{E}_z(\vec{r}_\perp,\omega) + \mathrm{H.\,c.} \right), \quad (1)$$

where $\hat{E}_z(\vec{r}_\perp,\omega)$ denotes the NIR field at the end facet of the detection crystal polarized along the crystallographic ⟨001⟩-axis and is given by the field of the probe signal, and $\hat{E}_x(\vec{r}_\perp,\omega)$ is the field along the ⟨110⟩, which is determined by the NIR vacuum fluctuations and the nonlinear field induced by the THz vacuum field and the source radiation. The influence of the wave-plate retardance γ is described by the following function:

$$P(\gamma) = \sqrt{-\cos(\gamma)} - \sqrt{2}\mathrm{i}\cos\left(\frac{\gamma}{2}\right). \quad (2)$$

Within the experiment, we measure two electro-optic signals, $\hat{S}^t_{\gamma_t}(0)$ and $\hat{S}^\tau_{\gamma_\tau}(\delta t)$, where δt denotes the delay between the two probe pulses. For vacuum fluctuations, both detection schemes employ a quarter-wave plate ($\gamma_t = \gamma_\tau = \frac{\pi}{2}$), and the experimentally observed correlation is given by:

$$G^{(1)}_{\mathrm{VF}}(\delta t) = \frac{\langle 0|\hat{S}^t_{\frac{\pi}{2}}(0), \hat{S}^\tau_{\frac{\pi}{2}}(\delta t)|0\rangle}{2\,C_{\mathrm{eo}}^2 N_t N_\tau}. \quad (3)$$

For source radiation, the detection configuration consists of a half-wave plate for the first probe ($\gamma_t = \pi$) and a quarter-wave plate for the second ($\gamma_\tau = \frac{\pi}{2}$), yielding

$$G^{(1)}_{\mathrm{SR}}(\delta t) = \frac{\langle 0|\hat{S}^t_{\pi}(0), \hat{S}^\tau_{\frac{\pi}{2}}(\delta t)|0\rangle}{2\,C_{\mathrm{eo}}^2 N_t N_\tau}. \quad (4)$$

Here, N_t and N_τ denote the total number of detected photons ($\langle\hat{N}_s\rangle + \langle\hat{N}_p\rangle$) in the two probe signals t and τ. The electro-optic coupling constant $C_{\mathrm{eo}} = \frac{n_p^3 r_{41}\omega_p}{2c} l$ depends on the probe refractive index $n_p = 2.85$ at frequency $\omega_p = 2\pi \times 375$ THz, the electro-optic coefficient of zinc telluride $r_{41} = 3.9$ pmV^{-1}, the speed of light c, and the length of the detection crystal $l = 1$ mm, and converts the unitless correlation of photon numbers into a field correlation as presented in Fig. 2.

Theoretical model

To model the experiments, we use the theoretical framework of two-beam electro-optical sampling developed in ref. 7. It builds on previous works in refs. 13,26.

Applying the paraxial wave-approximation to all involved near-infrared fields but not the THz field[26], the electro-optic sampling signal with two quarter-wave plates reads[7]

$$G_{\mathrm{vac}}(\delta t) = \int_{V_c} \mathrm{d}^3 r \int \mathrm{d}t \int_{V_c} \mathrm{d}^3 r' \int \mathrm{d}t' L_1(\mathbf{r},t) L_2(\mathbf{r}',t') C(\mathbf{r},\mathbf{r}',t,t'). \quad (5)$$

With a quarter-wave plate for pulse one and a half-wave plate for pulse two it is in turn given by:

$$G_s(\delta t) = -\frac{\hbar}{2}\int_{V_C} \mathrm{d}^3 r \int \mathrm{d}t \int_{V_C} \mathrm{d}^3 r' \int \mathrm{d}t' L_1(\mathbf{r},t) L_2(\mathbf{r}',t') R(\mathbf{r},\mathbf{r}',t,t'). \tag{6}$$

Here, V_C is the volume of the nonlinear crystal and $L_1(\mathbf{r},t) = L_2(\mathbf{r} - \delta r \mathbf{e}_x, t - \delta t) = (2/\pi)^{3/2} \mathrm{e}^{-2(t-n_g z/c)^2/\tau_o^2} \mathrm{e}^{-2r_\parallel^2/w^2}$ [$\mathbf{r}_\parallel = (r_x, r_z)$, w: beams waist, τ_o: pulse duration, n_g: group refractive index in the near-infrared] are the Gaussian-shaped pulse envelopes of the two laser pulses. The response and correlation functions of the x-polarized THz field $\widehat{E}(\mathbf{r},t)$ inside the crystal accounting for dispersion and absorption are defined in the main text and can be evaluated using macroscopic quantum electrodynamics[27]. In frequency domain, they are given by $R(\mathbf{r},\mathbf{r}',\Omega) = \mu_0 \Omega^2 D(\mathbf{r},\mathbf{r}',\Omega)/(2\pi)$ and $C(\mathbf{r},\mathbf{r}',\Omega) = \hbar\mu_0 \mathrm{sgn}[\Omega]\Omega^2 \mathrm{Im}[D(\mathbf{r},\mathbf{r}',\Omega)]/(2\pi)$, respectively, with the xx component of the dyadic Green tensor of the vector Helmholtz equation D and the vacuum permeability μ_0.

We further follow ref. 7 to numerically evaluate Eqs. (5) and (6). The resulting frequency domain correlation signals defined by $G_i(\Omega) = (1/2\pi)\int_{-\infty}^{\infty} \mathrm{d}\delta t G_i(\delta t) \mathrm{e}^{\mathrm{i}\Omega\delta t}$ are shown in Fig. 2c. We used the following parameters in the simulation, which, except for n_g, have all been independently determined experimentally: pulse duration τ_{FWHM} = 110 fs, Gaussian beam width w = 10 µm, beam separation δr = 50 µm, and the group refractive index in the near-infrared n_g = 3.18. The refractive index in the THz is given by:

$$n(\Omega) = \sqrt{\epsilon_\infty \left(1 + \frac{(\hbar\omega_{\mathrm{LO}})^2 - (\hbar\omega_{\mathrm{TO}})^2}{(\hbar\omega_{\mathrm{TO}})^2 - (\hbar\Omega)^2 - \mathrm{i}\hbar\Omega\gamma}\right)}, \tag{7}$$

with all parameters as measured in ref. 28 for a ZnTe crystal at temperature T = 300 K, except that we use ϵ_∞ = 7.38 to match the real part of the refractive index measured at T = 4 K reported in ref. 12.

Note that the influence of the phonon resonance at 5.3 THz starts to increase above 4 THz, which in turn leads to an enhancement of the local (longitudinal) field fluctuations[26], but also to an increase in absorption. As a result of the absorption and the longitudinal nature of these fluctuations, the spatial range of correlations in the quantum vacuum as well as the propagation of source radiation from one beam to the other are delimited, and the detected electro-optic correlation in Fig. 2c vanishes in that frequency range. This means that the bandwidth of our detection is actually not limited by the bandwidth of the laser pulse but by the onset of increasing losses due to the phonon resonance above ≈4 THz, and makes it crucial that our theoretical framework accounts for absorption via a complex permittivity.

The fluctuation dissipation theorem implies $C(\mathbf{r},\mathbf{r}',\Omega) = \hbar\,\mathrm{sgn}(\Omega)\,\mathrm{Im}[R(\mathbf{r},\mathbf{r}',\Omega)]$, leading to the relation between $G_s(\Omega)$ and $G_{\mathrm{vac}}(\Omega)$ discussed in the main text[7].

Data availability

The experimental data generated in this study have been deposited in the ETHZ Research Collection database under accession code (DOI) https://doi.org/10.3929/ethz-b-000723846[29].

References

1. Biswas, A. K. et al. Virtual photons and causality in the dynamics of a pair of two-level atoms. *Phys. Rev. A* **42**, 7 (1990).
2. Valentini, A. Non-local correlations in quantum electrodynamics. *Phys. Lett. A* **153**, 321–325 (1991).
3. Milonni, P. W., James, D. F. & Fearn, H. Photodetection and causality in quantum optics. *Phys. Rev. A* **52**, 2 (1995).
4. Pozas-Kerstjens, A. & Martín-Martínez, E. Harvesting correlations from the quantum vacuum. *Phys. Rev. D Part. Fields Gravit. Cosmol.* **92**, 064042 (2015).
5. Milonni, P. W. *The Quantum Vacuum: An Introduction to Quantum Electrodynamics* ISBN 9780080571492 (Academic Press, 2013).
6. Dalibard, J., Dupont-Roc, J. & Cohen-Tannoudji, C. Vacuum fluctuations and radiation reaction: identification of their respective contributions. *J. Phys.* **43**, 1617–1638 (1982).
7. Lindel, F., Herter, A. M., Faist, J. & Buhmann, S. Y. Probing vacuum field fluctuations and source radiation separately in space and time. *Phys. Rev. Res.* **5**, 043207 (2023).
8. Tjoa, E. & Martín-Martínez, E. When entanglement harvesting is not really harvesting. *Phys. Rev. D* **104**, 12 (2021).
9. Milonni, P. W., Ackerhalt, J. R. & Smith, W. A. Interpretation of radiative corrections in spontaneous emission. *Phys. Rev. Lett.* **31**, 958 (1973).
10. Senitzky, I. R. Radiation-reaction and vacuum-field effects in Heisenberg-picture quantum electrodynamics. *Phys. Rev. Lett.* **31**, 955 (1973).
11. Sabín, C., Del Rey, M., García-Ripoll, J. J. & León, J. Fermi problem with artificial atoms in circuit QED. *Phys. Rev. Lett.* **107**, 150402 (2011).
12. Settembrini, F. et al. Detection of quantum-vacuum field correlations outside the light cone. *Nat. Commun.* **13**, 1 (2022).
13. Moskalenko, A. S. et al. Paraxial theory of direct electro-optic sampling of the quantum vacuum. *Phys. Rev. Lett.* **115**, 26 (2015).
14. Riek, C. et al. Direct sampling of electric-field vacuum fluctuations. *Science* **350**, 6259 (2015).
15. Benea-Chelmus, I. C., Settembrini, F. F., Scalari, G. & Faist, J. Electric field correlation measurements on the electromagnetic vacuum state. *Nature* **568**, 7751 (2019).
16. Onoe, S. et al. Realizing a rapidly switched Unruh-DeWitt detector through electro-optic sampling of the electromagnetic vacuum. *Phys. Rev. D* **105**, 056023 (2022).
17. Pottier, N. & Mauger, A. Quantum fluctuation-dissipation theorem: a time-domain formulation. *Phys. A Stat. Mech. Appl.* **291**, 327–344 (2001).
18. Settembrini, F. F., Herter, A. & Faist, J. Third order nonlinear correlation of the electromagnetic vacuum at near-infrared frequencies. *N. J. Phys.* **26**, 043017 (2024).
19. Galiffi, E. et al. Photonics of time-varying media. *Adv. Photonics* **4**, 014002 (2022).
20. Moore, G. T. Quantum theory of the electromagnetic field in a variable–length one–dimensional cavity. *J. Math. Phys.* **11**, 2679–2691 (1970).
21. Fuentes-Schuller, I. & Mann, R. B. Alice falls into a black hole: entanglement in noninertial frames. *Phys. Rev. Lett.* **95**, 120404 (2005).
22. Mann, R. B. & Ralph, T. C. Relativistic quantum information. *Class. Quantum Grav.* **29**, 220301 (2012).
23. Lindel, F. et al. Entanglement harvesting from electromagnetic quantum fields. *Phys. Rev. A* **110**, 022414 (2024).
24. Kizmann, M. et al. Subcycle squeezing of light from a time flow perspective. *Nat. Phys.* **15**, 960–966 (2019).
25. Philbin, T. G. et al. Fiber-optical analog of the event horizon. *Science* **319**, 1367–1370 (2008).
26. Lindel, F., Bennett, R. & Buhmann, S. Y. Theory of polaritonic quantum-vacuum detection. *Phys. Rev. A* **102**, 041701 (2020).
27. Scheel, S. & Buhmann, S. Y. Macroscopic QED—concepts and applications. *Acta Phys. Slov.* **58**, 675–809 (2008).
28. Leitenstorfer, A. et al. Detectors and sources for ultrabroadband electro-optic sampling: experiment and theory. *Appl. Phys. Lett.* **74**, 1516–1518 (1999).
29. Herter, A. et al. Data Collection of current publication. ETHZ Research Collection. https://doi.org/10.3929/ethz-b-000723846 (2025).

Acknowledgements
A.H. acknowledges financial support from the Swiss National Science Foundation, grand number 207795. F.L. acknowledges support by the Spanish Ministry for Science and Innovation-Agencia Estatal de Investigación (AEI) through Grant No. EUR2023-143478, as well as by the Quantum Center Research Fellowship and the Dr. Alfred and Flora Spälti Fonds.

Author contributions
J.F., A.H., and F.L. conceived the idea for the experiment and its theoretical interpretation. A.H. and L.G. modified the experimental setup. A.H. conducted the measurements. The data analysis was primarily performed by A.H., and their results were interpreted by A.H., F.L., and J.F. The theoretical framework was developed by F.L. and S.Y.B. F.L. performed the numerical simulations. J.F. was the scientific supervisor of this work. The manuscript was written through contributions from all authors. All authors have given approval to the final version of the manuscript.

Competing interests
The authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-69142-4.

Correspondence and requests for materials should be addressed to Alexa Herter or Jérôme Faist.

Peer review information *Nature Communications* thanks Andrey Moskalenko and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026